

Question	Answer	Marks	Guidance
8(a)(i)	photoelectric effect	B1	Allow 'photoelectric emission' or 'photoemission'
8(a)(ii)	<u>minimum</u> frequency needed to cause emission (of electrons)	B1	
8(a)(iii)	<u>Any three points from:</u> - electrons need a minimum energy to leave the surface - photon absorbed by electron must have enough energy for electron to leave - photons are packets of energy proportional to frequency - minimum photon energy means minimum frequency (of radiation)	B3	Allow 'photon absorbed by electron must have energy at least equal to the work function (energy)' or reverse argument Allow 'quantum' for 'packet'
8(b)(i)	$hc / \lambda = \phi + E_K$	C1	Allow $hf = \phi + E_K$ and $f = c / \lambda$ Allow $hc / \lambda = hc / \lambda_0 + E_K$
	$h \times 3.00 \times 10^8 \times [(1 / 178 \times 10^{-9}) - (1 / 254 \times 10^{-9})]$ $= (5.27 - 1.91) \times 10^{-19}$	C1	
	$h = 6.66 \times 10^{-34} \text{ J s}$	A1	Must be given to 3 significant figures. AFC. No credit is implied by $6.63 \times 10^{-34} \text{ J s}$.

Question	Answer	Marks	Guidance
8(b)(ii)	$\phi = [(6.66 \times 10^{-34} \times 3.00 \times 10^8) / (178 \times 10^{-9})]$ $- (5.27 \times 10^{-19})$ or $\phi = [(6.66 \times 10^{-34} \times 3.00 \times 10^8) / (254 \times 10^{-9})]$ $- (1.91 \times 10^{-19})$ (= 5.95×10^{-19} J)	C1	<u>Independent C1 marks.</u> Allow '6.63' for '6.66'. Allow '6.63' for '6.66'. Possible ECF from 8(b)(i) . 5.9 or 6.0×10^{-19} J implies C1.
	division by 1.60×10^{-19}	C1	
	$\phi = (5.95 \times 10^{-19}) / (1.60 \times 10^{-19})$ = 3.7 eV	A1	Correct to 2 significant figures. AFC. Allow answers to greater than 2 SF for this question if answer is correct when rounded to 2 SF. Both sets of data and both values of h give 3.7.